Monthly Assessment - October 2026

Artificial Intelligence - Grade X

Set: B	Maximum Marks: 20	Suggested Time: 45 minutes

ANSWER KEY & MARKING GUIDELINES

SECTION A — MCQs (10 × 1 = 10 marks)

Q1. B) Textual data

Q2. C) Sentiment analysis

Q3. C) Lexical analysis

Q4. B) Stop words

Q5. C) Establishing relationships between preceding and succeeding sentences

Q6. B) Analysing a customer review for sentiment

Q7. A) Tokenisation

Q8. B) Stemming

Q9. B) Make differently cased forms more consistent

Q10. B) Interacts with users through text or speech

SECTION B — SUBJECTIVE (10 marks)

Q11. State the significance of NLP and mention one real-world use. (2 marks)

Suggested answer: NLP helps computers process and respond to human language. It is used in voice assistants, chatbots, translation and review analysis.

Q12. What are stop words? Give two examples. (2 marks)

Suggested answer: Stop words are common words that may add limited value to a text-analysis task and may be filtered out. Examples: “the” and “and”.

Q13. Explain two practical applications of NLP. (2 marks)

Suggested answer: Voice assistants use NLP to understand requests; sentiment analysis classifies opinions in reviews as positive, negative or neutral.

Q17. Explain how Bag of Words converts text into numerical features, using two short documents. (4 marks)

Suggested answer / marking points: Example D1 “data science needs data”; D2 “science uses models”. Vocabulary: data, science, needs, uses, models. D1 vector [2,1,1,0,0]; D2 vector [0,1,0,1,1]. Each vector records the vocabulary-word counts in that document.

Accept equivalent correct answers. Award partial credit for relevant concepts and correct intermediate steps.